

Direct Numerical Simulations of Mixed-Phase Cloud Glaciation Caused by Entrainment and Mixing

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Key Points:

- For the same initial ice mass, glaciation is enhanced if ice crystals are smaller and many compared to larger and few.
- A higher level of flow turbulence and mixing increases the rate of glaciation for any ice number density.
- A hexagonal column shape promotes glaciation while a hexagonal plate shape suppresses glaciation compared to spherical ice crystals.

Abstract

The glaciation of mixed-phase clouds is investigated at the microscale by use of direct numerical simulations (DNS) and Lagrangian particle tracking. The dry air is entrained into the cloud volume and subsequent turbulent mixing is simulated. The effects of flow turbulence intensity, ice number density, and ice crystal geometry on glaciation are studied. Our particle-based DNS model shows that high turbulence has an expediting effect on ice growth and glaciation. Glaciation is enhanced if the ice mass is distributed among smaller and many ice crystals compared to larger and few ice crystals. A hexagonal column shape promotes ice growth and glaciation compared to a sphere with the same volume. However, a hexagonal plate shape suppresses glaciation relative to spherical ice crystals for the simulated parameters. Neither ice number density nor ice crystal geometry has any impact on cloud droplet evaporation.

1 Introduction

Mixed-phase clouds are found at temperatures between -40° to 0° C (Korolev et al., 2017). In this temperature range, the saturation vapor pressure and supersaturation are higher with respect to water than they are with respect to ice at a given temperature. Thus, the air can be supersaturated with respect to ice but subsaturated with respect to water. That will cause ice crystals to grow but cloud droplets to shrink in a mixed-phase cloud. The increase of ice mass at the expense of liquid mass is known as the Wegener-Bergeron-Findeisen (WBF) process (Storelvmo & Tan, 2015) and is commonly referred to as glaciation. Mixed-phase clouds can fully glaciate or persist in a mixed-phase state (Morrison et al., 2012). The glaciation or persistence of mixed-phase clouds is a source of uncertainty in climate models and needs to be investigated at the microscale.

The WBF process was studied analytically by Korolev (2007) who showed that the mass of water vapor might increase or decrease during glaciation depending on the mass and number densities of cloud droplets and ice crystals. The analytical investigation by Pinksly et al. (2015) suggested that the growth of ice mass is dependent on ambient temperature. Mixed-phase clouds have also been studied numerically. Wang et al. (2024) studied the glaciation process using a large eddy simulation (LES) model. The authors reported that the rate of glaciation is higher if mixed-phase clouds grow in a well-mixed environment. Yang et al. (2015) applied LES to examine the growth of ice crystals in a mixed-phase cloud and found that a fraction of ice crystals gets trapped in turbulent eddies and grow or sublimate faster depending on the local ice supersaturation. LES was used by Ovchinnikov et al. (2014) who pointed out that ice number density influences mixed-phase cloud growth and the partitioning between cloud droplets and ice crystals needs accurate representation in numerical models. Chen et al. (2023) investigated the evolution of mixed-phase clouds using direct numerical simulations (DNS) and suggested that the size distributions of cloud droplets are more sensitive to turbulence than those of ice crystals and the initial liquid mass has to be large enough for the mixed-phase condition to persist.

The geometry of ice crystals is important, particularly at the microscale. Using a theoretical approach, Chen & Lamb (1994) showed that the aspect ratio ice crystals changes as the ice mass grows with time. Bailey & Hallett (2004) experimentally demonstrated that the complexity of ice crystal shape increases if the supersaturation with respect to ice is higher. Sheridan et al. (2009) suggested that ice growth is isotropic if ice crystals are larger initially but is anisotropic for smaller ice crystals. Sulia & Harrington (2011) found that glaciation can be under-predicted by nearly an order of magnitude if ice crystals are assumed to be spherical instead of the precise shape. Hence, further investigation is needed to determine the effects of ice crystal shape on glaciation.

In the present study, we implement DNS with Lagrangian tracking of cloud droplets and ice crystals to study the glaciation of mixed-phase clouds. We modify our previous DNS model (Sharfuddin et al., 2025) to solve the ice phase equations. The objective is to better understand ice microphysics and its interaction with ambient turbulence. There are two novel aspects of this study. First, we simulate the entrainment of dry air into the mixed-phase cloud and subsequent turbulent mixing with variation of flow turbulence intensity and ice number density. Second, we compare the growth profiles for different ice crystal geometries: sphere, hexagonal column, and hexagonal plate.

2 Governing Equations

The turbulent flow of air is governed by the incompressible Navier-Stokes equations:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho_a} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{F}_c, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (2)$$

Here, $\mathbf{u}$ is the instantaneous velocity vector, ν is the kinematic viscosity, p is the instantaneous pressure and ρ_a is the density of air. $\mathbf{F}_c$ is an external force to sustain turbulence as described in Sharfuddin et al. (2025). The transport equations for the thermodynamic fields of temperature (T) and water vapor mixing ratio (q_v) are:

$$\frac{\partial T}{\partial t} + (\mathbf{u} \cdot \nabla) T = \frac{L_w}{c_p} C_w + \frac{L_{ice}}{c_p} C_{ice} + \mu_T \nabla^2 T, \quad (3)$$

$$\frac{\partial q_v}{\partial t} + (\mathbf{u} \cdot \nabla) q_v = -(C_w + C_{ice}) + \mu_v \nabla^2 q_v. \quad (4)$$

Above, L_w and L_{ice} are specific latent heats of condensation and deposition, respectively. C_w is the time rate of mass exchange between vapor and liquid while C_{ice} denotes the same between vapor and ice. Also, μ_T and μ_v are molecular diffusivities of temperature and vapor, respectively. The environmental supersaturation (S_e) depends on the T and q_v , and is expressed as:

$$S_e = \frac{q_v}{q_{v,s}} - 1 = \frac{q_v}{621.97 \frac{p_{sat}}{p - p_{sat}}} - 1, \quad (5)$$

where $q_{v,s}$ is the saturation vapor mixing ratio and p_{sat} is the saturation vapor pressure. p_{sat} can be calculated with respect to water and ice (Huang, 2018) as:

$$p_{sat} = \begin{cases} p_{sat,w} = 611.2 \times \exp\left(\frac{17.67(T - 273.15)}{T - 29.65}\right) \\ p_{sat,ice} = 611.2 \times \exp\left(\frac{22.46(T - 273.15)}{T - 0.53}\right) \end{cases}. \quad (6)$$

The cloud droplets and ice crystals are tracked individually in Lagrangian fashion. The equations for the position ($\mathbf{X}$) and velocity ($\mathbf{V}$) vectors can be written as:

$$\frac{d\mathbf{X}_i(t)}{dt} = \mathbf{V}_i(t), \quad \frac{d\mathbf{V}_i(t)}{dt} = \frac{1}{(\tau_p)_i} [\mathbf{u}_i(\mathbf{X}, t) - \mathbf{V}_i(t)] + \mathbf{g}, \quad (\tau_p)_i = \frac{2\rho_i(r_i)^2}{9\rho_a\nu}. \quad (7)$$

where the subscript ' i ' denotes the i -th particle (cloud droplet or ice crystal). $\mathbf{u}_i$ is the instantaneous fluid velocity vector interpolated at the position of particle i and τ_p is the particle response time.

Conveniently, the S_e and particle radius (r) can be written as:

$$S_e = \begin{cases} S_{e,w} & \text{if } p_{sat} = p_{sat,w} \\ S_{e,ice} & \text{if } p_{sat} = p_{sat,ice} \end{cases}, \quad r = \begin{cases} r_w & \text{if } S_e = S_{e,w} \\ r_{ice} & \text{if } S_e = S_{e,ice} \end{cases}. \quad (8)$$

87 The growth equations for cloud droplets and ice crystals can be written as:

$$\frac{d(r_w(t))_i}{dt} = \left(\frac{G_w}{r_w(t)} S_{e,w}(\mathbf{X}, t) \right)_i, \quad \frac{d(r_{ice}(t))_i}{dt} = \left(\frac{G_{ice}}{r_{ice}(t)} S_{e,ice}(\mathbf{X}, t) \right)_i. \quad (9)$$

88 G_w and G_{ice} are expressed as:

$$G_w = \frac{1}{\frac{L_w \rho_w}{k_T T} \left(\frac{L_w}{R_v T} - 1 \right) + \frac{\rho_w R_v T}{\mu_v p_{sat,w}}}, \quad G_{ice} = \frac{1}{\frac{L_{ice} \rho_{ice}}{k_T T} \left(\frac{L_{ice}}{R_v T} - 1 \right) + \frac{\rho_{ice} R_v T}{\mu_v f p_{sat,ice}}}. \quad (10)$$

89 where k_T is the thermal conductivity in air, R_v is the specific gas constant for water vapor, and f
90 is the ice crystal shape factor. The f is unity for spherical ice crystals. For hexagonal ice crystals,
91 the f is found as: (Westbrook et al., 2008)

$$f = 0.58(1 + 0.95A^{0.75})(b/r_{ice}), \quad A = \frac{h}{2b}. \quad (11)$$

92 Above, b is the side length and h is the height of a regular hexagonal prism, and A is the aspect
93 ratio. The shape is hexagonal column if $A > 1$ and hexagonal plate if $A < 1$. r_{ice} is the equivalent
94 radius of hexagonal ice crystals and is related to b and h as:

$$\frac{4}{3}\pi r_{ice}^3 = \frac{3\sqrt{3}}{2}b^2h, \quad r_{ice} = 0.8527(b^2h)^{1/3}, \quad b = 0.93\left(\frac{r_{ice}}{A^{1/3}}\right). \quad (12)$$

95 We neglect processes like ice nucleation, secondary ice production, accretion, aggregation, and
96 ventilation in our model and focus on diffusion-limited ice growth only. With the foregoing, C_w
97 and C_{ice} can be determined as:

$$C_w(\mathbf{x}, t) = \frac{4\pi\rho_w}{\rho_a a^3} \sum_{i=1}^{n_w} \left(G_w r_w(t) S_{e,w}(\mathbf{X}, t) \right)_i, \quad (13)$$

$$C_{ice}(\mathbf{x}, t) = \frac{4\pi\rho_{ice}}{\rho_a a^3} \sum_{i=1}^{n_{ice}} \left(G_{ice} r_{ice}(t) S_{e,ice}(\mathbf{X}, t) \right)_i. \quad (14)$$

98 where ρ_w and ρ_{ice} are liquid and ice densities, respectively. n_w is the number of cloud droplets in
99 a grid cell with a volume of a^3 and n_{ice} is the same for ice crystals.

100 3 Numerical Details

101 To vary the turbulent kinetic energy (TKE) dissipation rate, we force different wavenumber
102 bands as discussed in Sharfuddin et al. (2025). Specifically, we use the wavenumber bands: $1 \leq$
103 $k_f/k_{min} \leq 3.0$ and $1 \leq k_f/k_{min} \leq 16.18$ to generate ‘low’ and ‘high’ turbulence levels, with
104 Kolmogorov length scales of 0.00092 m and 0.00226 m, respectively. Note that k_f is the forced
105 wavenumber and $k_{min} = 2\pi/0.512 = 12.27$ 1/m is the minimum wavenumber.

106 The physical model simulated is that of a cubic domain inside which turbulence, cloud
107 droplets, and ice crystals interact. The domain volume is 0.512^3 m³ and the boundary conditions
108 are triply periodic. At the start of simulations, the domain is divided equally into cloudy and clear-
109 air regions. Cloud droplets and ice crystals are randomly placed in the cloudy region. A slab-like
110 configuration is assumed so that the initial temperature and vapor mixing ratio change sharply at

the cloud/clear-air interface (Sharfuddin et al., 2025). This setup allows us to simulate entrainment and mixing (Gao et al., 2018). We specify the initial vapor mixing ratio field as:

$$q_v(z, t = 0) = \begin{cases} q_v^{max}, & d - L \times 0.5 \times 0.5 \leq z < d + L \times 0.5 \times 0.5 \\ q_{v,e}, & \text{elsewhere} \end{cases} \quad (15)$$

Above, $q_v^{max} = 2.367 \text{ g/kg}$, $q_{v,e} = 1.90 \text{ g/kg}$, and $d \equiv L/2$ is the width of the cloud slab. q_v^{max} corresponds to the cloudy region and $q_{v,e}$ represents the dry (subsaturated) region. The temperature field is initialized as:

$$T(t = 0) = \langle T(t = 0) \rangle - 0.608 \langle T(t = 0) \rangle (q_v(z, t = 0) - \langle q_v(t = 0) \rangle). \quad (16)$$

We simulate six cases: L-H, H-H, L-L, H-L, H-H-C, and H-H-P. The details are provided in Table I which also lists the values of relevant parameters. Note that higher ice number density corresponds to smaller ice crystal radius and the vice versa so that the initial ice mass is constant for all the cases. The ice crystal shape is hexagonal column for Case H-H-C, hexagonal plate for Case H-H-P, and spherical otherwise. For hexagonal column, the initial values of f , b , and h are 1.148 , $5.14 \mu\text{m}$, and $25.69 \mu\text{m}$, respectively. The corresponding values for hexagonal plate are 1.082 , $9.46 \mu\text{m}$, and $7.57 \mu\text{m}$. The initial surface area of an ice crystal are $707 \mu\text{m}^2$, $930 \mu\text{m}^2$, and $896 \mu\text{m}^2$ for Cases H-H, H-H-C, and H-H-P, respectively while the volume is $1767 \mu\text{m}^3$ for the three cases. The mean temperature is $\sim 260 \text{ K}$ (Table 1) initially and does not change significantly due to smaller domain size and shorter simulation time in our DNS study. Hexagonal plate ice crystals are most common at this temperature (Bailey & Hallett, 2009) but hexagonal column shape is possible depending on the $S_{e,ice}$. The assumption of spherical ice crystals is rather idealistic but necessary for comparison with past studies. The initial cloud droplet number density is 59.6 cm^{-3} and is much larger than the ice number density (Korolev et al., 2017). The number of grid points is 256^3 and the grid resolution is 2 mm for all eight cases.

Table 1

Details of the test cases

Case	TKE dissipation rate (m^2/s^3)	Ice number intensity (cm^{-3})	Initial ice crystal radius (μm)	Initial cloud droplet radius (μm)	Aspect ratio (h/b) of ice crystals
L-H	0.00037	0.011	7.5	7.5	N/A
H-H	0.01388	0.011	7.5	7.5	N/A
L-L	0.00037	0.0011	16.16	7.5	N/A
H-L	0.01388	0.0011	16.16	7.5	N/A
H-H-C	0.01388	0.011	7.5	7.5	2.5
H-H-P	0.01388	0.011	7.5	7.5	0.4

Case	Comments
L-H	low (L) turbulence, high (H) ice number density, spherical
H-H	high (H) turbulence, high (H) ice number density, spherical
L-L	low (L) turbulence, low (L) ice number density, spherical
H-L	high (H) turbulence, low (L) ice number density, spherical

H-H-C high (H) turbulence, high (H) ice number density, hexagonal column (C)
H-H-P high (H) turbulence, high (H) ice number density, hexagonal plate (P)

Parameters and their values

Symbol	Value	Unit	Symbol	Value	Unit
L_w	2.50×10^6	J/kg	c_p	1005.0	J/kg/K
L_{ice}	2.83×10^6	J/kg	ρ_a	0.7674	kg/m ³
ρ_w	1000	kg/m ³	ρ_{ice}	917	kg/m ³
$p(t = 0)$	57160	N/m ²	$\langle T(t = 0) \rangle$	259.53	K
R_v	461.5	J/kg/K	k_T	0.0238	W/m/K
μ	1.64×10^{-5}	m ² /s	$\mu_T = \mu_v$	2.2×10^{-5}	m ² /s

4 Results and Discussion

The mean supersaturation is negative with respect to water but positive with respect to ice at the initial time. Therefore, ice crystals grow but cloud droplets evaporate, and the mixed-phase cloud glaciates. The effects of flow turbulence level, ice number density, and ice crystal geometry on thermodynamics and microphysics are discussed.

Figures 1(a)-(b) show the temporal evolution of the mean supersaturation fields for the four cases: L-H, H-H, L-L, and H-L. Liquid water is converted into water vapor during evaporation of cloud droplets and ice crystals grow by absorbing that water vapor. Since the number density and mass of cloud droplets exceed those of ice crystals in our model, more water vapor is produced by evaporation than consumed by deposition to ice crystal surfaces. So, there is a net increase of water vapor mixing ratio (q_v). The $\langle S_{e,w} \rangle$ and $\langle S_{e,ice} \rangle$ increase because they are directly proportional to the $\langle q_v \rangle$. Flow turbulence accelerates entrainment and mixing which leads to faster evaporation of cloud droplets (Hoffmann, 2020). Hence, the $\langle S_{e,w} \rangle$ and $\langle S_{e,ice} \rangle$ increase faster for high turbulence cases (H-H and H-L) compared to low turbulence cases (L-H and L-L). In contrast, the DNS study of Chen et al. (2023) found no impact of flow turbulence on the mean supersaturation fields. This is because the dry (subsaturated) air is entrained into the cloud (supersaturated) in our model unlike that of Chen et al. (2023) which assumed a uniform distribution of the mean supersaturation fields at the initial time. It is also evident that the mean profiles in Figures 1(a)-(b) are identical for Cases H-H and H-L and for Cases L-H and L-L. Thus, ice number density or ice crystal radius has no influence on the $\langle S_{e,w} \rangle$ and $\langle S_{e,ice} \rangle$.

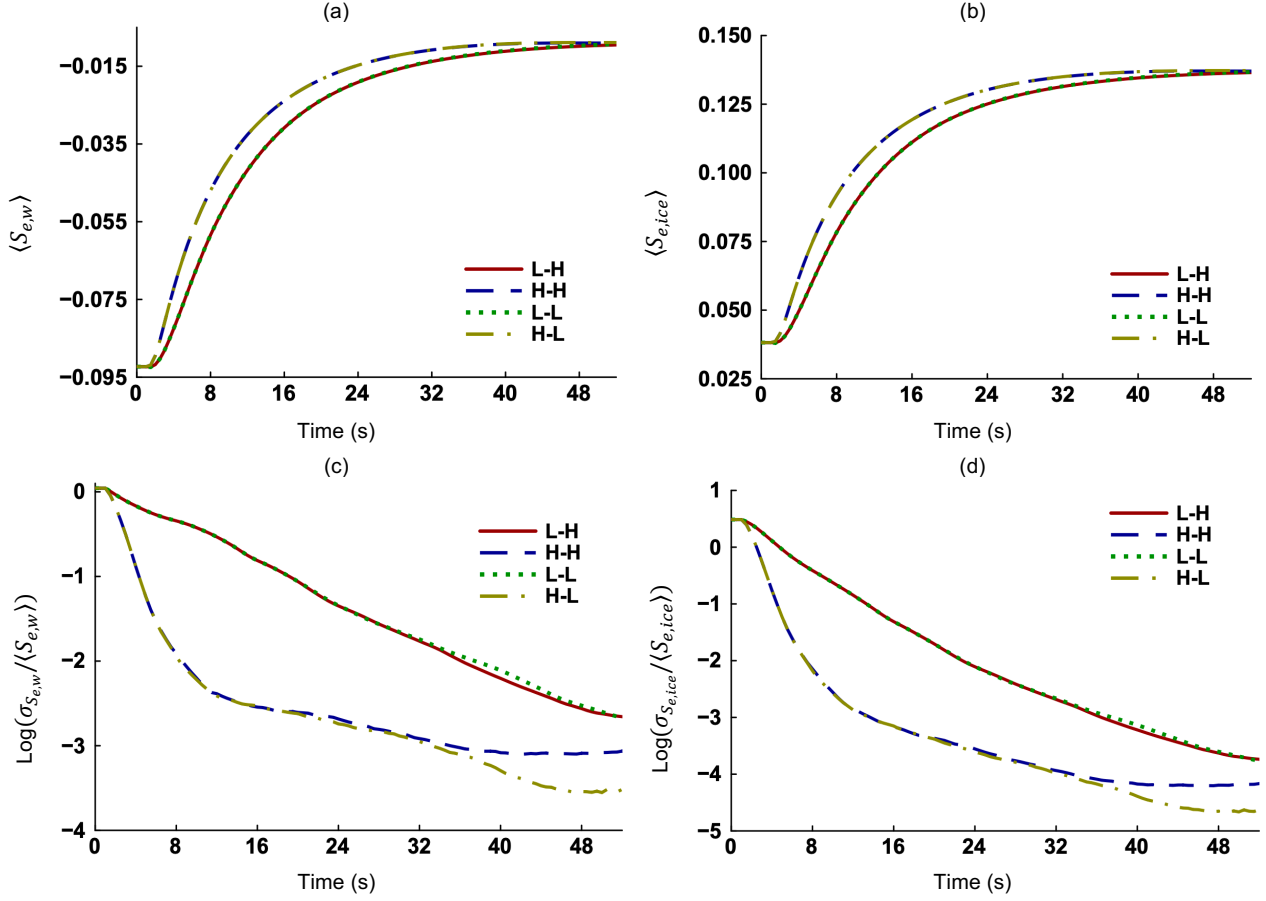


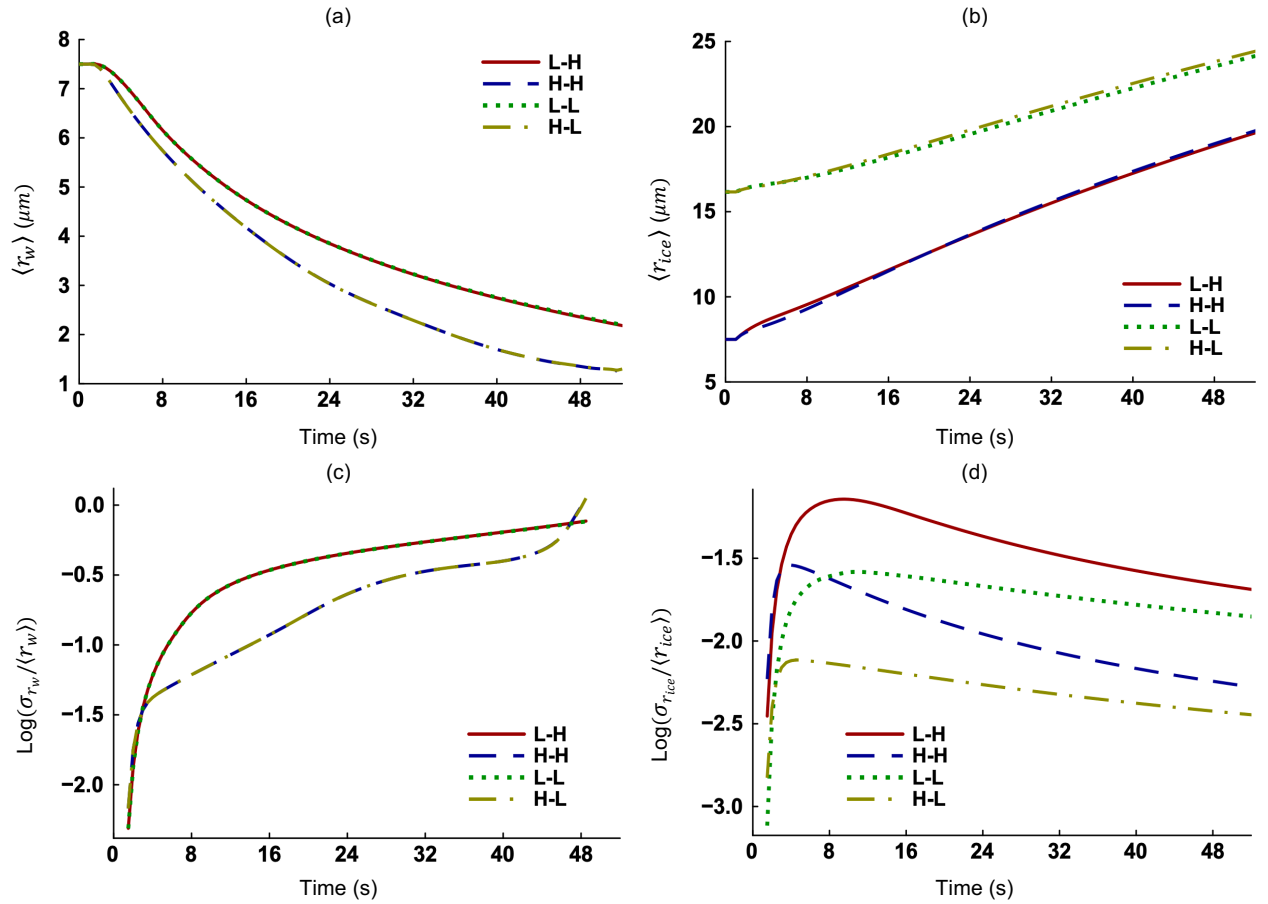
Figure 1. Time evolution of the (a) mean environmental supersaturation with respect to water ($\langle S_{e,w} \rangle$), (b) mean environmental supersaturation with respect to ice ($\langle S_{e,ice} \rangle$), (c) relative dispersion of supersaturation with respect to water ($\sigma_{S_{e,w}}/\langle S_{e,w} \rangle$), and (d) relative dispersion of supersaturation with respect to ice ($\sigma_{S_{e,ice}}/\langle S_{e,ice} \rangle$), for Cases L-H, H-H, L-L, and H-L.

The time evolution of the relative dispersion, which is the standard deviation divided by the mean, is shown in Figures 1(c)-(d). We see that $\sigma_{S_{e,w}}/\langle S_{e,w} \rangle$ and $\sigma_{S_{e,ice}}/\langle S_{e,ice} \rangle$ are maximum at the start. According to our initial configuration, the dry air and cloudy air regions have distinct supersaturation levels (Equation (15)) and are separated by a sharp interface. Turbulent mixing causes the sharp change in supersaturation at the interface to propagate throughout the domain and homogenizes the supersaturation fields. Hence, the $\sigma_{S_{e,w}}/\langle S_{e,w} \rangle$ and $\sigma_{S_{e,ice}}/\langle S_{e,ice} \rangle$ decrease with time. The higher the turbulence level (Cases H-L and H-H), the quicker the reduction of standard deviation and relative dispersion (Sharfuddin et al., 2025).

Consistent with Figures 1(a)-(b), stronger evaporation for Cases H-H and H-L reduces the mean cloud droplet radius ($\langle r_w \rangle$) compared to the other two cases (Figure 2(a)). Evaporation slows down at late times as the environment becomes less subsaturated (negative $S_{e,w}$). The mean radius of ice crystals ($\langle r_{ice} \rangle$) increases almost linearly for the four cases (Figure 2(b)). This observation agrees with the finding of Chen et al. (2023). Evaporation of cloud droplets produces more water vapor than ice growth can consume. Ice crystals keep growing at the same pace at early and late times due to the availability of water vapor. It appears that $\langle r_{ice} \rangle$ increases slightly faster for Case H-L compared to Case L-L, indicating that high turbulence has an expediting effect on ice growth.

The same is true for Cases H-H and L-H at late times. Stronger turbulence more evenly distributes the water vapor produced by evaporation throughout the domain. So, the chances for water vapor molecules to get attached to nearby ice crystal surfaces increase, leading to faster ice growth.

Figures 2(c)-(d) show the time evolution of the relative dispersions of cloud droplet and ice crystal radii, respectively. We see that $\sigma_{r_w}/\langle r_w \rangle$ and $\sigma_{r_{ice}}/\langle r_{ice} \rangle$ increase very fast at early times as the initially monodisperse cloud droplets and ice crystals are exposed to turbulent mixing. After that, the relative dispersions depend on diffusional growth. The $\sigma_{r_w}/\langle r_w \rangle$ continues to increase but at a slower rate (Figure 2(c)) while the $\sigma_{r_{ice}}/\langle r_{ice} \rangle$ decreases after reaching a peak (Figure 2(d)) for all four cases. The diffusional growth theory (Equation (9)) suggests that particle growth rate slows down as its radius increases and the vice versa. Since r_w becomes smaller and r_{ice} becomes larger with time, the growth rate increases for the former but decreases for the latter. The variation due to diffusional growth in a turbulent environment increases and decreases similarly. We notice in Figure 2(d) that high flow turbulence has a retarding effect on the relative dispersion of ice crystal radii for any ice number density (Cases H-L vs L-L and H-H vs L-H). For similar levels of turbulence, high ice number density has an expediting effect on the $\sigma_{r_{ice}}/\langle r_{ice} \rangle$ (Cases H-H vs H-L and L-H vs L-L). The relative dispersion of cloud droplet radii is not affected by ice number density, but stronger flow turbulence tends to retard the $\sigma_{r_w}/\langle r_w \rangle$.



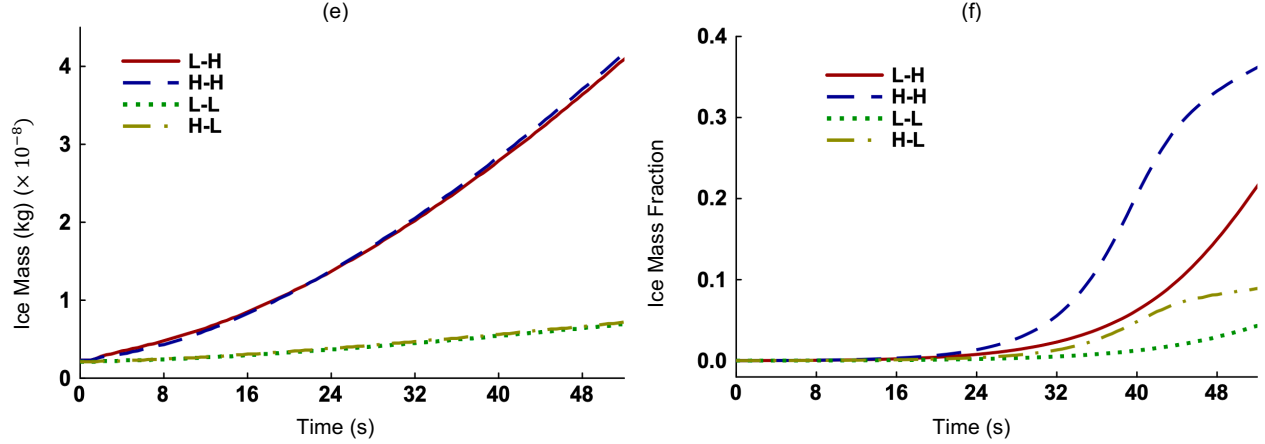
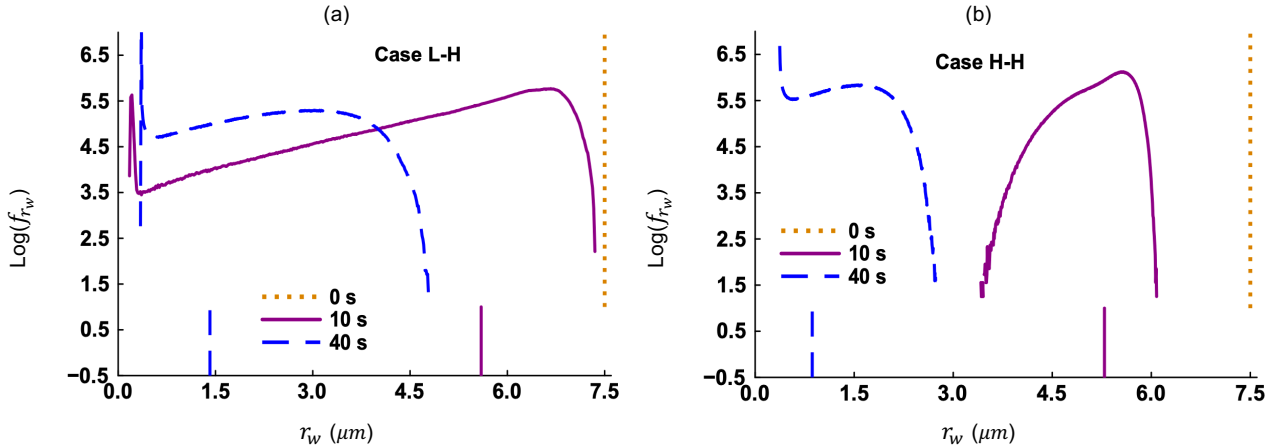


Figure 2. Time evolution of the (a) mean radius of cloud droplets ($\langle r_w \rangle$), (b) mean radius of ice crystals ($\langle r_{ice} \rangle$), (c) relative dispersion of cloud droplet radii ($\sigma_{r_w}/\langle r_w \rangle$), (d) relative dispersion of ice crystal radii ($\sigma_{r_{ice}}/\langle r_{ice} \rangle$), (e) ice mass, and (f) ice mass fraction, for Cases L-H, H-H, L-L, and H-L.

The initial mass of ice crystals is the same for the four cases. As expected during glaciation, the ice mass increases with time (Figure 2(e)). The increase is more significant when the ice mass is distributed among smaller and many ice crystals (Cases H-H and L-H) compared to larger and few ice crystals (Cases H-L and L-L). More ice crystals translate into a larger surface area to which water vapor can deposit (Korolev & Issac, 2003), leading to faster ice growth. We find that a higher flow turbulence level marginally increases the growth rate of ice mass. The ice mass fraction, or the ratio of ice mass and total mass of liquid and ice, increases most for Case H-H, followed by Cases L-H, H-L, and L-L, in order (Figure 2(f)). As high ice number density and high turbulence intensity separately increase the ice mass fraction, their combination leads to maximum glaciation (Case H-H). Similarly, low ice number density together with weaker turbulence causes minimum glaciation (Case L-L).



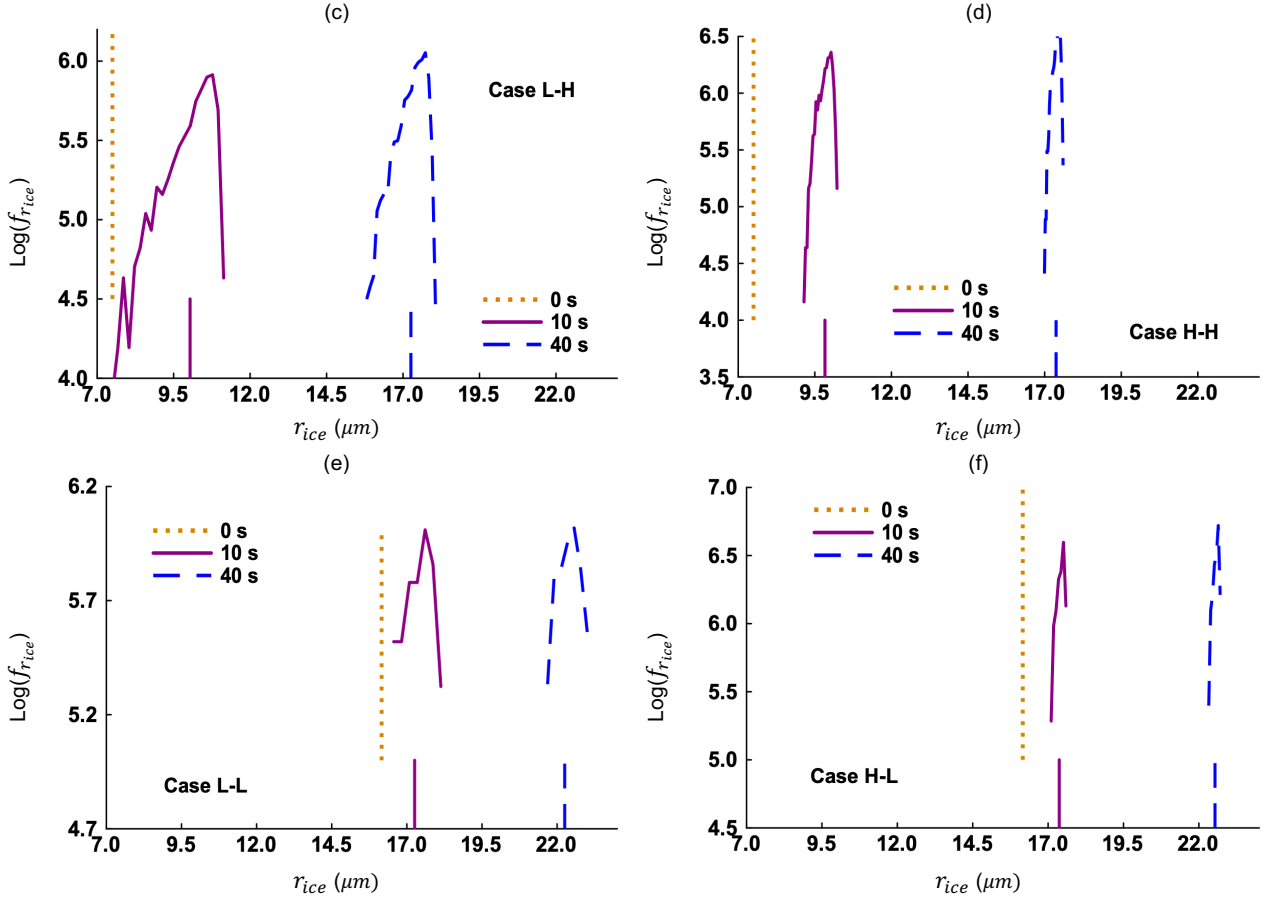


Figure 3. Log PDFs of cloud droplet radii (f_{r_w}) for Cases (a) L-H and (b) H-H. Log PDFs of ice crystal radii ($f_{r_{ice}}$) for Cases (c) L-H, (d) H-H, (e) L-L, and (f) H-L. Each plot shows the PDFs at 0 s, 10 s, and 40 s. The vertical line segments at the bottom indicate the mean values.

The probability density functions (PDFs) of cloud droplet radii (f_{r_w}) are shown in Figures 3(a)-(b) and those of ice crystal radii ($f_{r_{ice}}$) are shown in Figures 3(c)-(f). It appears that the f_{r_w} broadens between 0 s and 10 s for Case L-H because of turbulent fluctuations. It narrows between 10 s and 40 s (Figure 3(a)) as partial evaporation reduces maximum droplet radius. The f_{r_w} appears to be bimodal, with one peak for very small cloud droplets and another for larger droplets. Whereas stronger mixing leads to full evaporation of more cloud droplets and makes the f_{r_w} narrower for Case H-H (Figure 3(b)) compared to Case L-H. We find that some ice crystals have hardly grown from their initial $7.5 \mu\text{m}$ size by 10 s for Case L-H (Figure 3(c)). On the other hand, no ice crystal is smaller than $9 \mu\text{m}$ for Case H-H at 10 s (Figure 3(d)) although the $\langle r_{ice} \rangle$ is smaller. Between Cases L-H and L-L, the $f_{r_{ice}}$ are narrower for the latter (Figure 3(e)) due to low ice number density. The same is true between Cases H-H and H-L (Figure 3(f)). Unlike experimental observations, the present DNS model can faithfully distinguish between cloud droplets and ice crystals even if they are small.

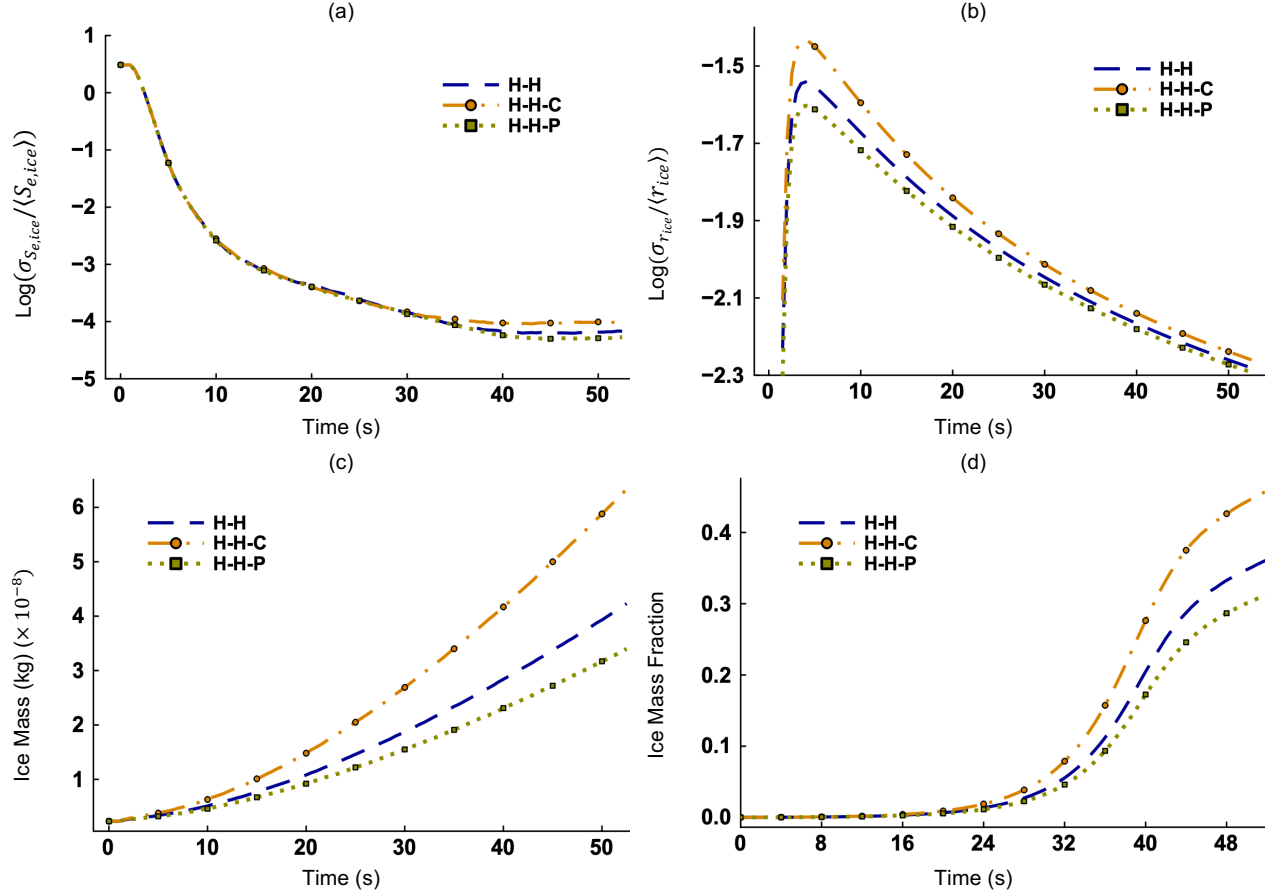


Figure 4. Time evolution of the (a) $\sigma_{S_{e,ice}} / \langle S_{e,ice} \rangle$, (b) $\sigma_{r_{ice}} / \langle r_{ice} \rangle$, (c) ice mass, and (d) ice mass fraction, for Cases H-H, H-H-C, and H-H-P.

Finally, we discuss the impact of ice crystal geometry on glaciation. It is assumed that the aspect ratios of hexagonal column ($A = 2.5$) and hexagonal plate ($A = 0.4$) do not change with time. The profiles of $\sigma_{S_{e,ice}} / \langle S_{e,ice} \rangle$ for the three cases (Figure 4(a)) differ at late times when they have decayed enough to be influenced by microphysical fluctuations. Figures 4(b)-(d) show that $\sigma_{r_{ice}} / \langle r_{ice} \rangle$ is larger and the increase of ice mass and ice mass fraction is faster if ice crystal shape is hexagonal column (H-H-C), followed by spherical (H-H), and hexagonal plate (H-H-P), in order. Thus, spherical ice crystals grow slower than hexagonal column ice crystals but faster than those with hexagonal plate geometry. Seo et al. (2012) and Shi et al. (2025) used molecular dynamics (MD) to simulate ice crystal growth at ~ 260 K and found that the basal faces of hexagonal ice crystals grow slower than the prism faces. The authors explained that the basal faces grow layer by layer and water molecules need to complete a full rearrangement before the next layer can grow. The prism faces grow through a collective three-dimensional mechanism, which can incorporate water molecules readily. Since the basal faces constitute a larger proportion of the total surface area for a hexagonal plate than for a hexagonal column, ice growth is slower for the former. So, the results of Seo et al. (2012), and Shi et al. (2025) agree with our model. Note that Equation (11) was derived by Westbrook et al. (2008) using a Monte Carlo (MC) approach and accounts for the attachment of water molecules to ice crystal surface. We also report that ice crystal shape has no impact on cloud droplet evaporation and influences the glaciation process by affecting ice growth only.

5 Concluding Remarks

We have studied the glaciation of turbulent mixed-phase clouds using a three-dimensional particle-based DNS model. Our results clarify several microphysical aspects of glaciation caused by entrainment and mixing. A high level of turbulence intensifies mixing between the cloud and environment and expedites glaciation. Glaciation is enhanced if the ice mass is distributed among smaller and many ice crystals compared to larger and few ice crystals. A hexagonal column shape promotes ice growth and glaciation while a hexagonal plate shape appears to suppress glaciation. The present letter also shows that a particle-based DNS model can address the uncertainties related to ice microphysics. We shall employ this model in the future to account for complex ice growth processes such as accretion and aggregation in mixed-phase clouds.

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Open Research

The data that support the findings of this work are available from the corresponding author upon reasonable request.

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